

Massimo Alioto

professor

基本信息

姓名: Massimo Alioto

所属院校: 新加坡国立大学

地区: Singapore

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

数字集成电路设计

个人简介

Massimo Alioto, born in Brescia in 1972, is a Professor at the ECE department of the National University of Singapore, leading the Green IC group. He holds an M.Sc. and Ph.D. in Electronic Engineering from the University of Catania. His research focuses on energy-efficient integrated circuits and systems, including ultra-low power circuits, IoT, wearables, and hardware security. Alioto has authored or co-authored over 330 papers and four books, and is an IEEE Fellow for contributions to energy-efficient VLSI circuits. He is the Editor-in-

Chief of IEEE Transactions on VLSI Systems and has been a Distinguished Lecturer for IEEE Solid-State Circuits Society and IEEE Circuits and Systems Society.

工作经历

National University of Singapore - Professor, Director of the Integrated Circuits and Embedded Systems area	2020-01-01 -
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National University of Singapore - Associate Professor, Director of the Integrated Circuits and Embedded Systems area	2013-07-01 - 2019-12-31
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Universita degli Studi di Siena Facolta di Ingegneria - Associate Professor	2006-11-01 - 2013-06-31
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Intel Corp Hillsboro - Visiting Scientist	2013-01-06 - 2013-03-28
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University of Michigan - Visiting Professor	2011-07-01 - 2013-01-01
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University of California Berkeley Research - Visiting Professor	2009-01-05 - 2011-06-30
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EPFL School of Engineering (STI) - Visiting Professor	2007-07-28 - 2007-08-31
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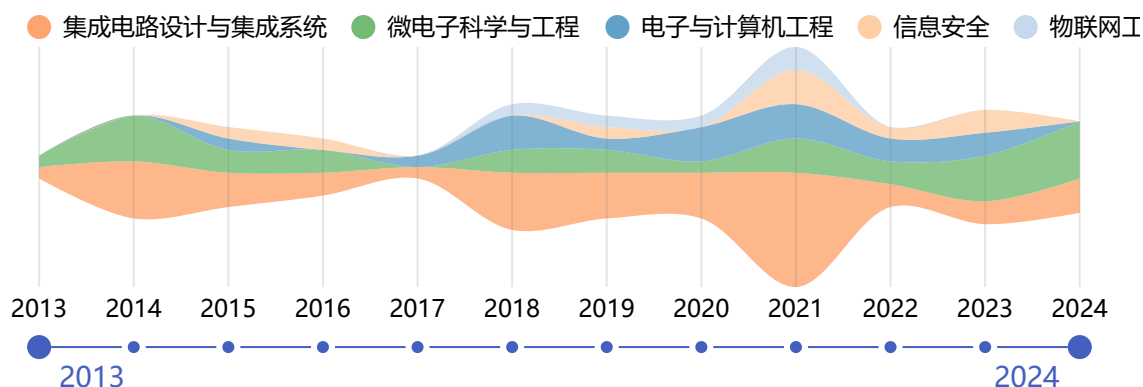
Universita degli Studi di Siena Facolta di Ingegneria - Assistant Professor	2002-10-01 - 2006-10-31
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教育经历

Universita degli Studi di Catania - PhD	1997-10-01 - 2001-02-27
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Universita degli Studi di Catania - MSc	1990-10-01 - 1997-07-25
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近年发表的论文



0.6-V, μ W- Power Four-Stage OTA With Minimal Components, and 100x Load Range

IEEE SOLID-STATE CIRCUITS LETTERS

Privitera, Marco; Grasso, Alfio Dario; Ballo, Andrea; Alioto, Massimo

被引用次数: 0

2024

Backscattered Software-Defined Radio for Flexible, Reusable and Upgradeable Transmitters with 34-58 pJ/bit Energy Across Common Standards in 180 nm

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Yang, Ruiyuan; Ahmed, Karim Ali; Alioto, Massimo

被引用次数: 0

2024

Event-Driven Voltage Reference with Bandgap-Class Temperature Coefficient Across-Corner with 100-pW Power at 0.8 V

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Fassio, Luigi; Hanh, Hoang Hong; Alioto, Massimo

被引用次数: 0

2024

A 15-nA quiescent current capacitor-less LDO for sub-1V μ W-powered fully-harvested systems

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Privitera, Marco; Ballo, Andrea; Grasso, Alfio Dario; Alioto, Massimo

被引用次数: 0

2024

Introduction to the Special Section on the 2023 IEEE International Solid-State Circuits Conference (ISSCC)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Mike Shuo-Wei; Sathe, Visvesh S.; Alioto, Massimo; Seo, Jae-Sun; Shiga, Hidehiro

被引用次数: 0

2024

Picowatt-Power Super-Cutoff Analog Building Blocks and 78-pW Battery-Less Wake-Up Receiver for Light-Harvested Near-Always-On Operation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Basu, Joydeep; Fassio, Luigi; Ali, Karim; Alioto, Massimo

被引用次数: 1

2024

Sub- μ W Battery-Less and Oscillator-Less Wi-Fi Backscattering Transmitter Reusing RF Signal for Harvesting, Communications, and Motion Detection

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Privitera, Marco; Ballo, Andrea; Ali, Karim; Grasso, Alfio Dario; Alioto, Massimo

被引用次数: 0

2024

Aggressive Design Reuse for Ubiquitous Zero-Trust Edge Security-From Physical Design to Machine-Learning-Based Hardware Patching

IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY

Alioto, Massimo

被引用次数: 5

2023

Capacitance-to-Digital Converter for Harvested Systems Down to 0.3 V With No Trimming, Reference, and Voltage Regulation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Aiello, Orazio; Crovetto, Paolo Stefano; Alioto, Massimo

被引用次数: 2

2023

Opening of the 2023 Editorial Year-This Coda as Prelude of Next TVLSI Cycle With Sustained Growth

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 0

2023

Guest Editorial IEEE 2022 European Solid-State Circuits Conference

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Nagata, Makoto; Alioto, Massimo

被引用次数: 0

2023

A 0.4V 12b Comparator Offset Injection Assisted SAR ADC achieving 0.425 fJ/conv-step

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Vohra, Japesh; Ahmed, Karim Ali; Alioto, Massimo

被引用次数: 1

2023

Single-Antenna Backscattered BLE5 Transmitter with up to 97m Range, 10.6 μ W Peak Power for Purely-Harvested Green Systems

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Ahmed, Karim Ali; Yang, Ruiyuan; Salamani, Praveenakumar; Rajanna, Viveka; Alioto, Massimo

被引用次数: 3

2023

Laser Voltage Probing Attack Detection With 100% Area/Time Coverage at Above/Below the Bandgap Wavelength and Fully-Automated Design

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Hui; Lin, Longyang; Fang, Qiang; Alioto, Massimo

被引用次数: 1

2023

Voltage Reference With Corner-Aware Replica Selection/Merging for 1.4-mV Accuracy in Harvested Systems Down to 3.9 pW, 0.2 V

IEEE ACCESS

Fassio, Luigi; Lin, Longyang; De Rose, Raffaele; Lanuzza, Marco; Crupi, Felice; Alioto, Massimo

被引用次数: 3

2023

State-of-the-Art Techniques for Hardware Security - From Design Methodologies to On-Chip Intelligent Sensors

2022 INTERNATIONAL CONFERENCE ON IC DESIGN AND TECHNOLOGY (ICICDT)

Alioto, Massimo

被引用次数: 0

2022

In-Memory Unified TRNG and Multi-Bit PUF for Ubiquitous Hardware Security

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Taneja, Sachin; Rajanna, Viveka Konandur; Alioto, Massimo

被引用次数: 28

2022

STT-BNN: A Novel STT-MRAM In-Memory Computing Macro for Binary Neural Networks

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Thi-Nhan Pham; Quang-Kien Trinh; Chang, Ik-Joon; Alioto, Massimo

被引用次数: 26

2022

Opening of the 2022 TVLSI Editorial Year-Connecting Trends From Society to VLSI Systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 1

2022

A 3.2-pW, 0.2-V Trimming-Less Voltage Reference with 1.4-mV Across-Wafer Total Accuracy

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Fassio, Luigi; Lin, Longyang; De Rose, Raffaele; Lanuzza, Marco; Crupi, Felice; Alioto, Massimo

被引用次数: 4

2021

A 300mV-Supply, Sub-nW-Power Digital-Based Operational Transconductance Amplifier

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Toledo, Pedro; Croveti, Paolo; Klimach, Hamilton; Bampi, Sergio; Aiello, Orazio; Alioto, Massimo

被引用次数: 17

2021

Rail-to-Rail Dynamic Voltage Comparator Scalable Down to pW-Range Power and 0.15-V Supply

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Aiello, Orazio; Croveti, Paolo; Toledo, Pedro; Alioto, Massimo

被引用次数: 33

2021

A Robust, High-Speed and Energy-Efficient Ultralow-Voltage Level Shifter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Fassio, Luigi; Settino, Francesco; Lin, Longyang; De Rose, Raffaele; Lanuzza, Marco; Crupi, Felice; Alioto, Massimo

被引用次数: 12

2021

TempDiff: Feature Map-Level CNN Sparsity Enhancement at Near-Zero Memory Overhead via Temporal Difference

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

De Alwis, Udari; Alioto, Massimo

被引用次数: 3

2021

SRAM with In-Memory Inference and 90% Bitline Activity Reduction for Always-On Sensing with 109 TOPS/mm² and 749-1,459 TOPS/W in 28nm

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Rajanna, Viveka Konandur; Taneja, Sachin; Alioto, Massimo

被引用次数: 4

2021

Second Quarter of the 2021 Editorial Year-A Year in Crescendo

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 1

2021

Opening of the 2021 Editorial Year-Overture for a New Year of Change

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 3

2021

From Less Batteries to Battery-Less Alert Systems with Wide Power Adaptation down to nWs-Toward a Smarter, Greener World

IEEE DESIGN & TEST

Alioto, Massimo

被引用次数: 20

2021

SRAM with In-Memory Inference and 90% Bitline Activity Reduction for Always-On Sensing with 109 TOPS/mm² and 749-1,459 TOPS/W in 28nm

IEEE 51ST EUROPEAN SOLID-STATE DEVICE RESEARCH CONFERENCE (ESSDERC 2021)

Rajanna, Viveka Konandur; Taneja, Sachin; Alioto, Massimo

被引用次数: 0

2021

Last-round and Joint First/Last-Round Power Analysis Attacks on PRESENT

PROCEEDINGS OF THE 2021 ASIAN HARDWARE ORIENTED SECURITY AND TRUST SYMPOSIUM (ASIANHOST)

Fang, Qiang; Alioto, Massimo

被引用次数: 0

2021

Trimming-Less Voltage Reference for Highly Uncertain Harvesting Down to 0.25 V, 5.4 pW

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fassio, Luigi; Lin, Longyang; De Rose, Raffaele; Lanuzza, Marco; Crupi, Felice; Alioto, Massimo

被引用次数: 26

2021

Design of Digital OTAs With Operation Down to 0.3 V and nW Power for Direct Harvesting

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Toledo, Pedro; Croveti, Paolo; Aiello, Orazio; Alioto, Massimo

被引用次数: 33

2021

A 0.6-to-1.8V CMOS Current Reference With Near-100% Power Utilization

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Fassio, Luigi; Lin, Longyang; De Rose, Raffaele; Lanuzza, Marco; Crupi, Felice; Alioto, Massimo

被引用次数: 11

2021

$\pm$ CIM SRAM for Signed In-Memory Broad-Purpose Computing From DSP to Neural Processing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jain, Saurabh; Lin, Longyang; Alioto, Massimo

被引用次数: 22

2021

On-Chip Links With Energy-Quality Tradeoff in Error-Resilient and Machine Learning Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Rajanna, Viveka Konandur; Alioto, Massimo

被引用次数: 1

2021

Fully Synthesizable Unified True Random Number Generator and Cryptographic Core

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Taneja, Sachin; Alioto, Massimo

被引用次数: 12

2021

PUF Architecture with Run-Time Adaptation for Resilient and Energy-Efficient Key Generation via Sensor Fusion

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Taneja, Sachin; Alioto, Massimo

被引用次数: 13

2021

Sub-nW Microcontroller With Dual-Mode Logic and Self-Startup for Battery-Indifferent Sensor Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Longyang; Jain, Saurabh; Alioto, Massimo

被引用次数: 15

2021

Low-Energy Voice Activity Detection via Energy-Quality Scaling From Data Conversion to Machine Learning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Teo, Jinq Horng; Cheng, Shuai; Alioto, Massimo

被引用次数: 11

2020

Fully Synthesizable Low-Area Analogue-to-Digital Converters With Minimal Design Effort Based on the Dyadic Digital Pulse Modulation

IEEE ACCESS

Aiello, Orazio; Crovetto, Paolo; Alioto, Massimo

被引用次数: 40

2020

Energy-Quality Scalable Memory-Frugal Feature Extraction for Always-On Deep Sub-mW Distributed Vision

IEEE ACCESS

Alvarez, Anastacia; Ponnusamy, Gopalakrishnan; Alioto, Massimo

被引用次数: 3

2020

Processor Energy-Performance Range Extension Beyond Voltage Scaling via Drop-In Methodologies

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jain, Saurabh; Lin, Longyang; Alioto, Massimo

被引用次数: 5

2020

Integrated Power Management for Battery-Indifferent Systems With Ultra-Wide Adaptation Down to nW

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Longyang; Jain, Saurabh; Alioto, Massimo

被引用次数: 18

2020

Editorial on the Opening of the New Editorial Year-The State of the IEEE Transactions on Very Large Scale Integration (VLSI) Systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 0

2020

Automated Design of Reconfigurable Microarchitectures for Accelerators Under Wide-Voltage Scaling

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Jain, Saurabh; Lin, Longyang; Alioto, Massimo

被引用次数: 2

2020

Editorial on the Conclusion of the 2020 Editorial Year-The Climactic Finale of a Peculiar Year

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 2

2020

Approximate Multipliers With Dynamic Truncation for Energy Reduction via Graceful Quality Degradation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Frustaci, Fabio; Perri, Stefania; Corsonello, Pasquale; Alioto, Massimo

Fully Synthesizable Low-Area Digital-to-Analog Converter With Graceful Degradation and Dynamic Power-Resolution Scaling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Aiello, Orazio; Crovetto, Paolo Stefano; Alioto, Massimo

被引用次数: 28

2019

Standard Cell-Based Ultra-Compact DACs in 40-nm CMOS

IEEE ACCESS

Aiello, Orazio; Crovetto, Paolo; Alioto, Massimo

被引用次数: 29

2019

TVLSI Positioning-Continuing and Accelerating an Upward Trajectory

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo

被引用次数: 6

2019

Energy-Quality Scalable Adders Based on Nonzeroing Bit Truncation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Frustaci, Fabio; Perri, Stefania; Corsonello, Pasquale; Alioto, Massimo

被引用次数: 21

2019

Token-Based Security for the Internet of Things With Dynamic Energy-Quality Tradeoff

IEEE INTERNET OF THINGS JOURNAL

Aman, Muhammad Naveed; Taneja, Sachin; Sikdar, Biplab; Chua, Kee Chaing; Alioto, Massimo

被引用次数: 41

2019

A pW-Power Hz-Range Oscillator Operating With a 0.3-1.8-V Unregulated Supply

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Aiello, Orazio; Crovetto, Paolo; Lin, Longyang; Alioto, Massimo

被引用次数: 41

2019

Reconfigurable Clock Networks for Wide Voltage Scaling

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Longyang; Jain, Saurabh; Alioto, Massimo

被引用次数: 2

2019

Dynamic Reference Voltage Sensing Scheme for Read Margin Improvement in STT-MRAMs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Quang-Kien Trinh; Ruocco, Sergio; Alioto, Massimo

被引用次数: 29

2018

Guest Editorial Special Issue on Selected Papers from PRIME 2017 and SMACD 2017

INTEGRATION-THE VLSI JOURNAL

Di Capua, Giulia; Horta, Nuno; Fernandez, Francisco V.; Dundar, Gunhan; Pennisi, Salvatore; Palumbo, Gaetano; Alioto, Massimo; Giustolisi, Gianluca

被引用次数: 0

2018

The Internet of Things on Its Edge Trends toward its tipping point

IEEE CONSUMER ELECTRONICS MAGAZINE

Alioto, Massimo; Shahghasemi, Mohsen

被引用次数: 70

2018

A Variation-Aware Timing Modeling Approach for Write Operation in Hybrid CMOS/STT-MTJ Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

De Rose, Raffaele; Lanuzza, Marco; Crupi, Felice; Siracusano, Giulio; Tomasello, Riccardo; Finocchio, Giovanni; Carpentieri, Mario; Alioto, Massimo

被引用次数: 44

2018

Time-Based Sensing for Reference-Less and Robust Read in STT-MRAM Memories

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Quang-Kien Trinh; Ruocco, Sergio; Alioto, Massimo

被引用次数: 26

2018

Fully Synthesizable PUF Featuring Hysteresis and Temperature Compensation for 3.2% Native BER and 1.02 fJ/b in 40 nm

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Taneja, Sachin; Alvarez, Anastacia B.; Alioto, Massimo

被引用次数: 78

2018

iRazor: Current-Based Error Detection and Correction Scheme for PVT Variation in 40-nm ARM Cortex-R4 Processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Yiqun; Khayat-zadeh, Mahmood; Yang, Kaiyuan; Saligane, Mehdi; Pinckney, Nathaniel; Alioto, Massimo; Blaauw, David; Sylvester, Dennis

被引用次数: 66

2018

Energy-Quality Scalable Circuits and Systems for Sensing and Computing: From Approximate to Communication-Inspired and Learning-Based

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Alioto, Massimo; De, Vivek; Marongiu, Andrea

被引用次数: 10

2018

Energy-Quality Scalable Integrated Circuits and Systems: Continuing Energy Scaling in the Twilight of Moore's Law

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Alioto, Massimo; De, Vivek; Marongiu, Andrea

被引用次数: 28

2018

Dynamically Adaptable Pipeline for Energy-Efficient Microarchitectures Under Wide Voltage Scaling

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jain, Saurabh; Lin, Longyang; Alioto, Massimo

被引用次数: 10

2018

Editorial

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chakrabarty, Krishnendu; Alioto, Massimo; Joshi, Rajiv

被引用次数: 0

2017

Design-Oriented Energy Models for Wide Voltage Scaling Down to the Minimum Energy Point

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jain, Saurabh; Lin, Longyang; Alioto, Massimo

被引用次数: 5

2017

Special Issue on Circuits and Systems for the Internet of Things-From Sensing to Sensemaking

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Sanchez-Sinencio, Edgar; Sangiovanni-Vincentelli, Alberto

被引用次数: 13

2017

A Novel Framework to Estimate the Path Delay Variability On the Back of an Envelope via the Fan-Out-of-4 Metric

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 22

2017

Voltage Scaled STT-MRAMs Towards Minimum-Energy Write Access

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Quang Kien Trinh; Ruocco, Sergio; Alioto, Massimo

被引用次数: 20

2016

Approximate SRAMs With Dynamic Energy-Quality Management

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Frustaci, Fabio; Blaauw, David; Sylvester, Dennis; Alioto, Massimo

被引用次数: 33

2016

Static Physically Unclonable Functions for Secure Chip Identification With 1.9-5.8% Native Bit Instability at 0.6-1 V and 15 fJ/bit in 65 nm

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Alvarez, Anastacia B.; Zhao, Wenfeng; Alioto, Massimo

被引用次数: 64

2016

Novel Boosted-Voltage Sensing Scheme for Variation-Resilient STT-MRAM Read

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Quang Kien Trinh; Ruocco, Sergio; Alioto, Massimo

被引用次数: 27

2016

Novel Self-Body-Biasing and Statistical Design for Near-Threshold Circuits With Ultra Energy-Efficient AES as Case Study

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhao, Wenfeng; Ha, Yajun; Alioto, Massimo

被引用次数: 28

2015

Variations in Nanometer CMOS Flip-Flops: Part I-Impact of Process Variations on Timing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Consoli, Elio; Palumbo, Gaetano

被引用次数: 75

2015

Variations in Nanometer CMOS Flip-Flops: Part II-Energy Variability and Impact of Other Sources of Variations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Consoli, Elio; Palumbo, Gaetano

被引用次数: 22

2015

SRAM for Error-Tolerant Applications With Dynamic Energy-Quality Management in 28 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Frustaci, Fabio; Khayatzadeh, Mahmood; Blaauw, David; Sylvester, Dennis; Alioto, Massimo

被引用次数: 59

2015

Analysis and Characterization of Variability in Subthreshold Source-Coupled Logic Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shoaran, Mahsa; Tajalli, Armin; Alioto, Massimo; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 11

2015

Novel Class of Energy-Efficient Very High-Speed Conditional Push-Pull Pulsed Latches

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Consoli, Elio; Palumbo, Gaetano; Rabaey, Jan M.; Alioto, Massimo

被引用次数: 12

2014

Tunnel FETs for Ultra-Low Voltage Digital VLSI Circuits: Part II-Evaluation at Circuit Level and Design Perspectives

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Alioto, Massimo; Esseni, David

被引用次数: 27

2014

Tunnel FETs for Ultra low Voltage Digital VLSI Circuits: Part I-Device-Circuit Interaction and Evaluation at Device Level

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Esseni, David; Guglielmini, Manuel; Kapidani, Bernard; Rollo, Tommaso; Alioto, Massimo

被引用次数: 56

2014

Effectiveness of Leakage Power Analysis Attacks on DPA-Resistant Logic Styles Under Process Variations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Bongiovanni, Simone; Djukanovic, Milena; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 55

2014

A 346 μm^2 VCO-Based, Reference-Free, Self-Timed Sensor Interface for Cubic-Millimeter Sensor Nodes in 28 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fick, Laura; Fick, David; Alioto, Massimo; Blaauw, David; Sylvester, Dennis

被引用次数: 23

2014

Introduction to the special issue on IEEE-NEWCAS 2012

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Desgreys, Patricia; Alioto, Massimo

被引用次数: 0

2013

EChO Reconfigurable Power Management Unit for Energy Reduction in Sleep-Active Transitions

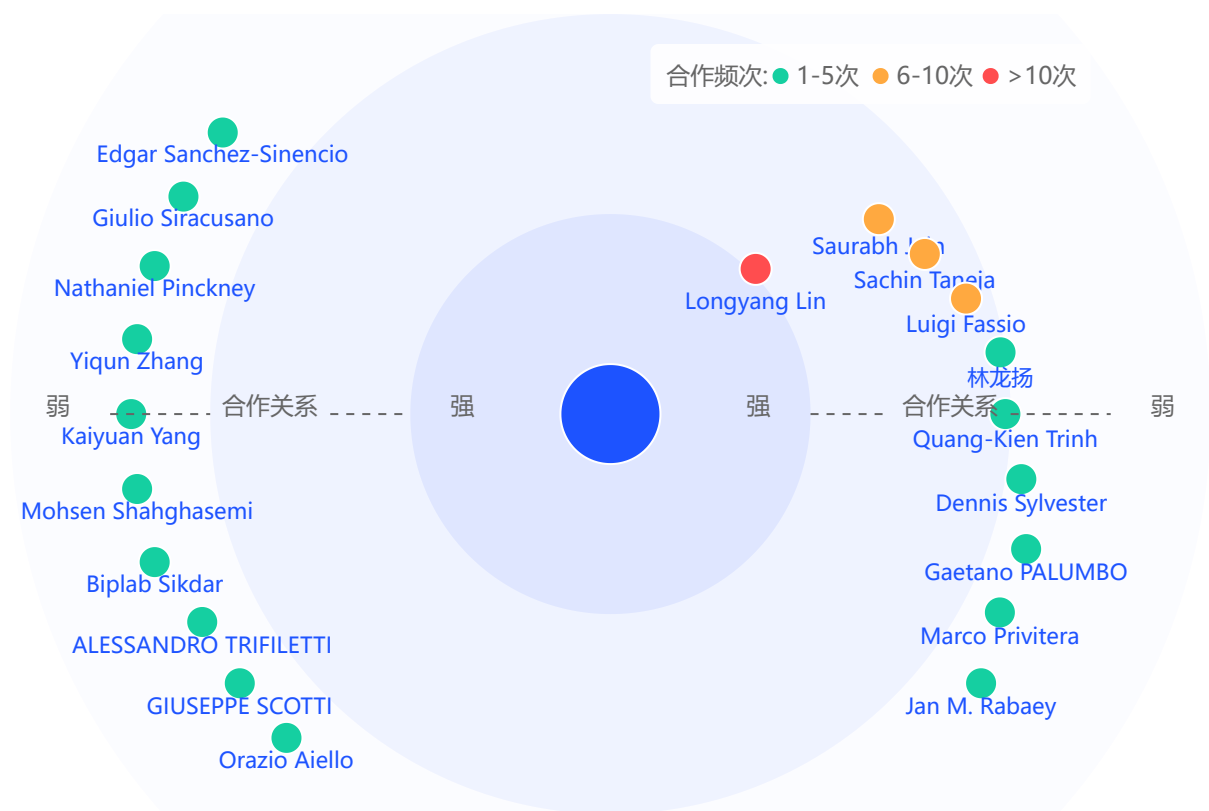
IEEE JOURNAL OF SOLID-STATE CIRCUITS

Alioto, Massimo; Consoli, Elio; Rabaey, Jan M.

被引用次数: 27

2013

学术合作



Longyang Lin

新加坡国立大学

合作次数: 14

Saurabh Jain

新加坡国立大学

合作次数: 8

Sachin Taneja

新加坡国立大学

合作次数: 7

Luigi Fassio

新加坡国立大学

合作次数: 6

林龙扬

南方科技大学

合作次数: 5

还有 22 个合作者